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Unlocking Sustainable and Innovative Hydraulic Fracturing Strategies for Deep Clastic Reservoir with Cost Optimization Opportunities

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Abstract

Hydraulic fracturing has traditionally been a key technology for enhancing productivity in low-permeability reservoirs. However, the cost, technical complexity, and risk associated with fracturing in deep, compartmentalized clastic formations have often limited its application, particularly in mature or brownfield developments. This paper outlines the successful execution of a cost-effective, sustainable, and scalable hydraulic fracturing campaign in Field-A, targeting a deep clastic reservoir with declining productivity and limited pressure support. Hydraulic Fracturing implemented in five key wells led to transformation from marginal or non-producing status to sustained, high-rate production. By retaining existing completions and carefully managing pressure envelopes, the project avoided two hoist-based workovers per well, saving approximately \$1.2 million each. An integrated approach involving well and performance reservoir performance diagnostics, real-time monitoring, risk assessments, and simulation tools (GOFFER, WellCat, CyberString) was employed to optimize fracture design under stringent pressure and equipment constraints, including the presence of TRSSSVs with limited pressure ratings.

Post-fracture results indicated dramatic increases in productivity—from 3 m³/d to over 340 m³/d in well A-6, and from shut-in to 250 m³/d in well A-20. The five-well program yielded an NPV of approximately \$158 million and a Unit Technical Cost (UTC) of \$1.04/bbl. The methodology not only delivered production uplift but also met sustainability metrics by avoiding ESPs, reducing emissions, and eliminating rig mobilizations. The results present a replicable model for operators facing similar geological and operational constraints.

Introduction

The continuous pursuit of optimizing hydrocarbon recovery from complex reservoirs underpins the global oil and gas industry's operational strategy, especially in mature and challenging plays. In this context, hydraulic fracturing has emerged as a transformative technology, particularly for low-permeability clastic formations that exhibit suboptimal productivity and declining pressure profiles. This paper presents a detailed case study of Field A—a deep clastic reservoir—where hydraulic fracturing was successfully

piloted and scaled up to overcome reservoir constraints, unlock latent production potential, and deliver significant economic value.

Historically, Field A faced numerous production challenges, including poor reservoir connectivity, inadequate pressure support, and inconsistent inflow performance across wells. Traditional stimulation and artificial lift techniques had reached their operational and economic limits. Therefore, an integrated and innovative hydraulic fracturing strategy was developed to target the reservoir with poor deliverability, with a strong focus on cost-efficiency, operational safety, and environmental sustainability.

The significance of this study lies not only in its technical accomplishments but also in the broader implications for hydraulic fracturing in similar environments. By leveraging existing well completions and minimizing invasive interventions such as hoist-based workovers, the project achieved a favorable balance between cost, risk, and performance. Advanced simulation and modeling tools were employed to optimize treatment designs within constrained wellhead and tubular pressure ratings, ensuring safe execution while enhancing productivity.

This paper outlines the planning, execution, and outcomes of hydraulic fracturing in five wells within Field A. It highlights the methodologies adopted, challenges encountered, and innovative solutions implemented to safely exceed conventional operational envelopes. In doing so, the project has set a new benchmark for sustainable and scalable hydraulic stimulation in deep clastic reservoirs, with early results indicating a step-change in production and a net present value (NPV) of approximately \$158 million.

Field Overview: A-Field

A-Field is located in the northern part of the basin in the middle of the Sultanate of Oman. It comprises two main producing reservoirs—Gharif and Al Khlata (AK)—both classified as deep, low-permeability clastic units situated roughly 1,500 meters beneath the Shuaiba carbonate reservoir.

First discovered in 1998, Field A is a medium sized asset within the region based on the Stock Tank Oil Initially In Place (STOIIP). The field development was executed in two major phases. To date, a total of 31 wells have been drilled.

Despite its sizeable STOIIP and comprehensive development efforts, the field has faced persistent production challenges, including reservoir pressure depletion, low matrix permeability, and early water breakthrough in certain flank areas. These challenges are compounded by several key subsurface uncertainties, most notably:

- The distribution and lateral connectivity of channel sands in the Gharif Formation, particularly at the flanks,
- The risk of encountering fluid contacts due to the shallow and variable oil-water contact (OWC) in some parts of the field.

Conventional production enhancement methods, including artificial lift and acid stimulation, have yielded limited success. As a result, several wells were classified as low-productivity or shut-in shortly after completion.

The relatively good wellbore conditions in many of these wells, coupled with availability of frac equipment due to downtime in other fields, presented an opportunity for field reactivation with optimized cost. This backdrop formed the foundation for selecting A-Field as a pilot area for a targeted hydraulic fracturing campaign aimed at:

- Enhancing productivity from previously marginal wells,
- Reactivating shut-in assets without workover, and
- Demonstrating the feasibility of cost-effective fracture stimulation in deep clastic reservoirs.

The case studies that follow describe in detail the planning, execution, and results of this initiative, which has since contributed to a step-change in field performance and laid the groundwork for future expansion of the fracturing program across the asset.

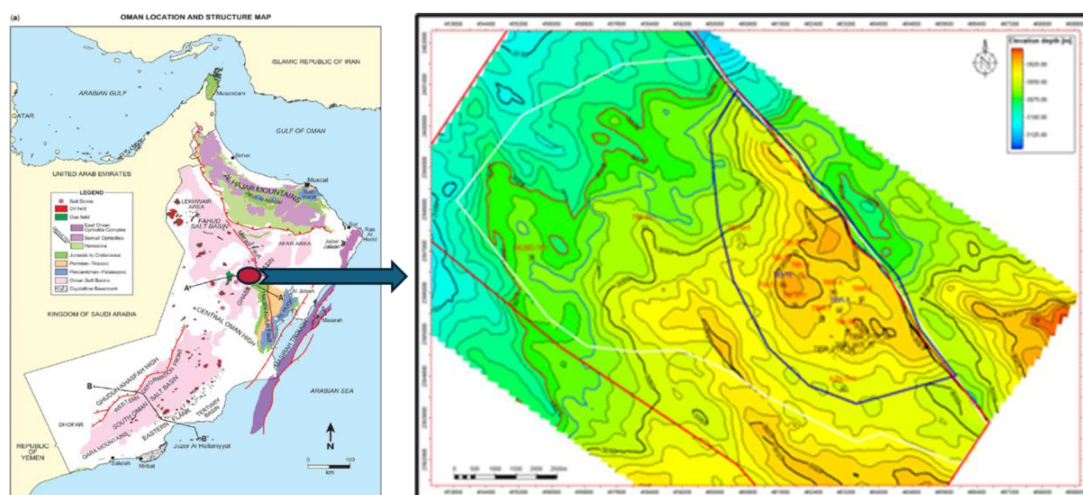


Figure 1—Field A Location and Subsurface Map.

Methodology

To address the production constraints in Field A's deep clastic reservoir, a comprehensive, multidisciplinary approach was adopted. The methodology encompassed three core stages: (1) detailed reservoir diagnostics and candidate selection, (2) fracture treatment design within existing operational constraints, and (3) execution and continuous optimization across the fracturing campaign.

Reservoir Characterization and Candidate Well Selection

The first phase focused on robust subsurface diagnostics to understand the root causes of poor well performance. This involved:

- **Reservoir Modeling and Petrophysical Analysis:** Core and log data were analyzed to characterize lithology, porosity, permeability, and net pay thickness. The reservoir exhibited typical deep clastic heterogeneity with compartmentalized sand bodies and variable pressure depletion patterns.
- **Production Performance Review:** Historical production trends and decline curves were reviewed. Candidate wells were selected based on low productivity index, poor reservoir connectivity, and low bottomhole pressure. Starting with low rate producing wells to avoid such high deferments until full strategy execution approved was considered.
- **Well Integrity and Completion Assessment:** An inventory of existing wells with completion configurations suitable for hydraulic fracturing was compiled. Wells with sufficient tubular integrity and pressure containment capability were prioritized to minimize cost and deferment.

Fracture Design under Operational Constraints

Given that the hydraulic fracturing was to be conducted using existing completions, the fracture design process had to be carefully engineered to operate within or safely exceed the design envelopes of surface and downhole equipment:

- **Integrated Risk Assessment (IRA):** Each treatment was subjected to a full-cycle IRA, involving input from production, drilling, completions, and HSE/ wells integrity teams. Risk mitigation plans were developed to manage pressure-related and operational hazards.

- **Fracturing Simulation and Modeling:** Fracture geometry and treatment parameters were modeled using advanced simulation software. Sensitivity studies were run on proppant type, fluid system, pump rates, and volumes to achieve optimal fracture conductivity without compromising well integrity. Due to pressure limitation identified from stress analysis, the treatment design was restricted in terms of proppant concentration and pumping rates. This was further optimized during execution based on mini frac and observation during main frac treatment. Further information is shared in the case studies listed in this paper.
- **Equipment Rating and Envelope Pushing:** Where required, wellhead and tubular ratings were carefully evaluated. In specific cases, fracturing designs slightly exceeded standard operating envelopes—supported by detailed engineering evaluations and real-time pressure monitoring to ensure safe execution.

Execution Strategy and Continuous Learning

The hydraulic fracturing program was rolled out in a pilot-to-field scale strategy, starting with single-well trials before full-field implementation:

- **Pump-Through Completions:** Frac jobs were performed via existing completion assemblies, avoiding costly hoist-based interventions, workovers, and recompletions.
- **Real-Time Monitoring and Diagnostics:** Surface and downhole data acquisition systems were employed to monitor treatment pressures, rates, and fluid efficiency in real time. Each job was followed by production testing and post-frac analysis.
- **Iterative Learning and Optimization:** Operational parameters and fracture designs were continuously refined based on actual job performance and early production results. A feedback loop ensured rapid adaptation and improved efficiency across subsequent wells.

This methodology enabled a low-cost, high-impact fracturing program that maintained strict safety and environmental standards while delivering transformative production enhancements.

Results and Discussion

The implementation of the hydraulic fracturing campaign in Field-A yielded a step-change in well performance and demonstrated the feasibility of deploying advanced stimulation strategies under constrained operating conditions. Five candidate wells were successfully fractured using the methodology outlined, resulting in significant gains across production, economics, and environmental performance.

Production Performance

Pre-frac production across the five wells averaged approximately 7 m³/d of net oil, with several wells exhibiting minimal or unstable inflow. Post-fracturing, the combined output increased to over 650m³/d, marking a nearly 93-fold improvement in daily production rates. Each fractured well contributed meaningfully to this uplift, with early-time production confirming the effectiveness of the fracture geometry and conductivity.

The initial production trends indicated:

- Inflow improvement in previously underperforming zones with stable production.
- Consistent post-frac productivity exceeding modeled expectations in three wells
- Favorable decline trends, suggesting sustained fracture conductivity and reservoir connectivity

This production transformation was achieved without any well workovers or major interventions, highlighting the value of the pump-through strategy and integrated engineering design.

Economic Impact

The hydraulic fracturing campaign delivered strong economic performance. The estimated net present value (NPV) of the five-well program was approximately \$158 million, driven by high early-time production, accelerated cash flow, and significant cost avoidance:

- Unit Technical Cost (UTC) was reduced to \$1.04/bbl, well below typical thresholds for deep clastic reservoirs.
- Hoist workover costs, averaging several million USD per well, were completely avoided.
- Re-hooking and deferment costs were minimized through strategic planning and streamlined execution.

The decision to retain and work through existing completions yielded substantial savings while enabling rapid mobilization and execution.

Operational and Technical Learnings

Several key learnings emerged during the course of the campaign:

- Operating Envelope Management: By applying rigorous modeling and risk assessment, it was possible to safely exceed standard pressure limitations in some wells. Real-time monitoring and conservative fracture design prevented equipment damage or well integrity compromise.
- Completion Compatibility: Not all completion strings were immediately suitable for frac treatments. The screening process identified limitations in a subset of wells, necessitating further design adjustments or exclusion from the initial campaign.
- Fracture Effectiveness: In low-permeability zones, the combination of optimized proppant selection and fluid systems resulted in long, highly conductive fractures. These were critical in overcoming formation damage and improving drawdown.
- Execution Efficiency: Continuous improvement from well to well led to a reduction in average job time and complexity. The final two wells in the campaign were executed with 25% less non-productive time compared to the first.

Environmental and Sustainability Gains

A significant outcome of the project was the reduction in GHG emissions and environmental impact. Key sustainability gains included:

- Avoidance of diesel-powered ESPs, which are typically used to manage low-pressure production
- Minimized surface footprint, as no workover rigs or heavy equipment mobilization was required
- Lower emissions intensity, attributed to reduced deferments and faster monetization of oil volumes

This integrated hydraulic fracturing approach aligns with broader industry goals to decarbonize upstream operations while maintaining financial viability.

Reviving Low-Productivity Wells through Hydraulic Fracturing: Case Studies from A-Field

Hydraulic fracturing has proven to be a key enabler in unlocking production potential from underperforming and shut-in wells. This paper presents some of the successful case studies from the A field—A-6, A-20 and new oil producers A-29 & A-30—demonstrating how data-driven fracture design, rigorous planning, and operational agility delivered significant production enhancements and cost savings.

Case-1: From Marginal Flow to High-Rate Producer

The well A-6 well had been producing at a marginal rate of only 3 m³/d (~18 bbl/day) for over 2.5 years. Following a comprehensive, integrated review of the Gharif formation across the nearby exploration wells, along with recent reservoir pressure monitoring, well A-6 was identified as the most suitable candidate for the first hydraulic fracturing treatment in the field based on its crestal location, its existing completion and good reservoir properties. Figure-2 shows well A-6 existing completion and its RDL. The well was completed across two intervals in UG-1 and UG-2 reservoirs.

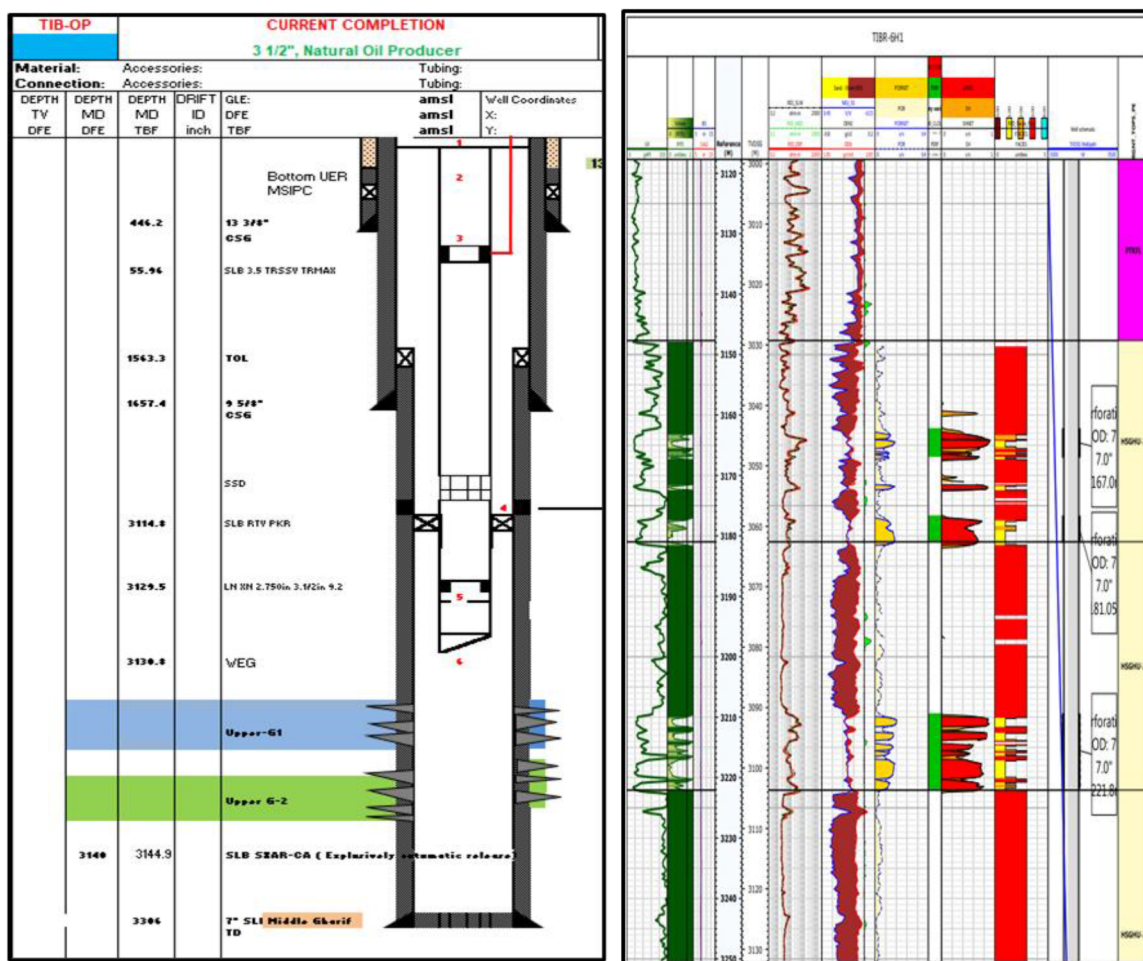


Figure 2—Well A-6 Existing completion and its RDL

The well's integrity was thoroughly verified using MAASP updates, SIT/WIT data, and well history, complemented by WellCat and CyberString modeling to confirm pressure envelope integrity and packer performance under fracturing conditions. Figure-3 shows WellCat modeling results that confirmed the suitability for frac with constrained pressure. The analysis showed that the maximum surface pressure during operation would be limited due to the packer rating limitations. To counter this, sensitivities were carried out with back pressure applied in the Annulus and that helped to increase the maximum surface injection pressure but was still limited to 6 ppg proppant concentration to stay within the envelope of the existing well completion.

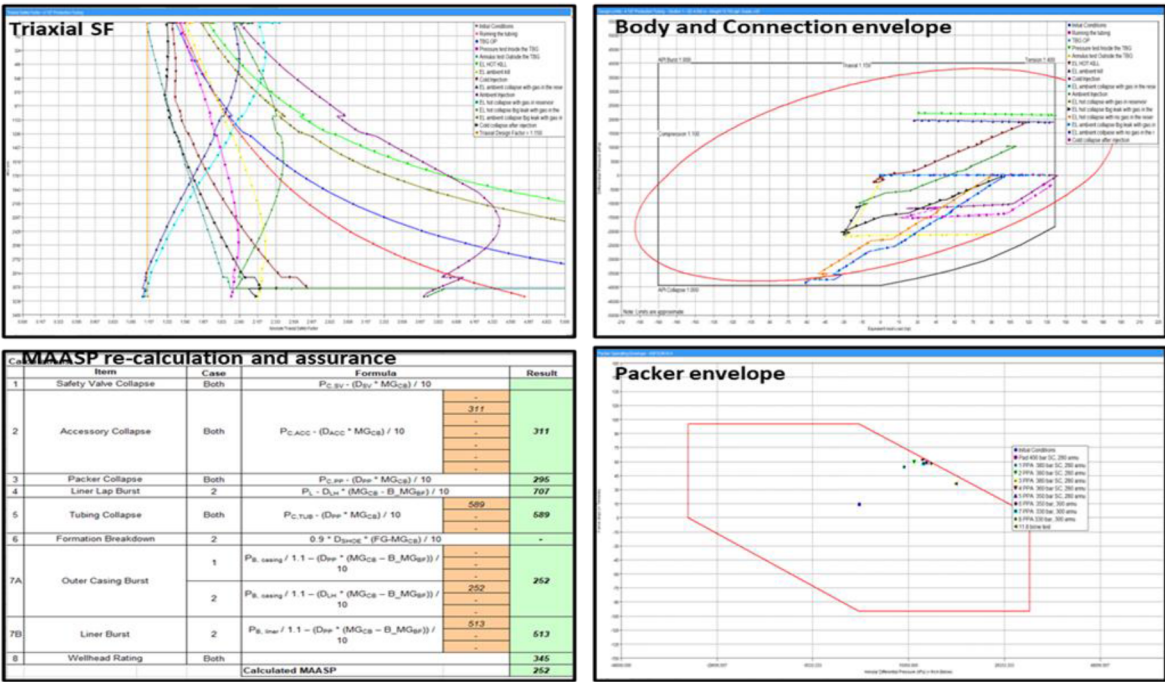


Figure 3—WellCat Analysis and Well/Completion Compliance to Frac (A-6)

A fracture design review using the GOFFER model evaluated three scenarios with varying injection pressures, rates, and proppant volumes. The plan was to frac both UG-1 and UG-2 zones in a single stage to avoid the complexity of a two-stage fracture, as both zones were already perforated and exhibited similar reservoir properties. It was acknowledged that controlling individual fracture growth in UG-1 and UG-2 would be challenging since they were fractured simultaneously through existing perforations.

Despite this, the decision to proceed with a single-stage frac was supported by the similar pressure profiles and reservoir characteristics of both zones, as well as modeling results indicating fracture propagation in both UG-1 and UG-2.

To validate this approach, GR tracers were deployed in one well and logged after the frac to assess proppant distribution between the two zones. The results showed nearly equal proppant distribution, reinforcing confidence in the single-stage strategy. Furthermore, production data from the initial wells confirmed the success of this approach.

The final design selected was based on pumping 100 metric tons (MT) of proppant at a rate of 4 m³/min based on the FCD and the concern of possible connection to the water injectors in the field. Figure-4 indicating Frac design selection.

Fracture	Final PPA	Treating Rate (m ³ /min)	Main Frac. Clean Volume (m ³)	Total Proppant (MT)	Gross Frac Length. M	Proppant Cutoff Length. M	Fracture Height. M	Average Prop Conc. kg/m ³ (>1lb/sqft eq.)	Conductivity 20/40 ISP mD.m	Conductivity 16/30 ISP mD.m	FCD 20/40 ISP	FCD 16/30 ISP
Grid Model from RDL												
Design 1 - 100 MT	8	4	416	100	788	144	12.97	4.9	1170	2030	81.25	140.97
Design 2 - 120 MT	8	4	474	120	800	200	12.97	4.9	1170	2030	58.50	101.50
Design 3 - 150 MT	8	4	563	150	800	264	12.97	4.9	1170	2030	44.32	76.89

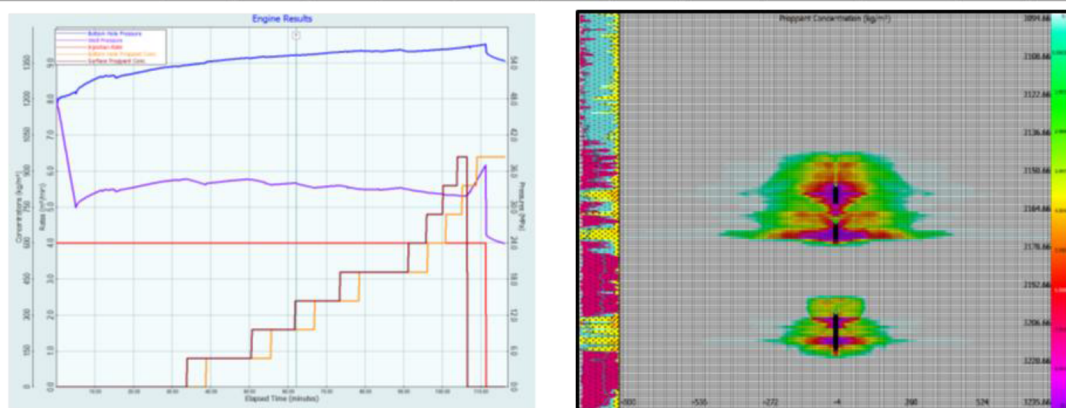


Figure 4—Designs Review & Selection (A-6)

Pre-fracturing diagnostics included a breakdown test, fluid efficiency test (FET), and step-rate test (SRT), which provided critical data for treatment refinement. Breakdown pressure was recorded at 25,374 kPa at 0.56 m³/min, with maximum pressure reaching 33,803 kPa and a fluid efficiency of 3.8%. Based on mini-frac results, a 20 m³ Pre-PAD stage using AF-35 linear gel was incorporated, and the PAD volume was adjusted to 5 m³. The main treatment was executed safely, with 84.6 tons of proppant pumped and 84.3 tons successfully placed in the formation. The main treatment was limited to 84.3 MT due to the maximum surface pressure as per WellCat simulations had been reached. Main treatment plot is shown in below figure (5).

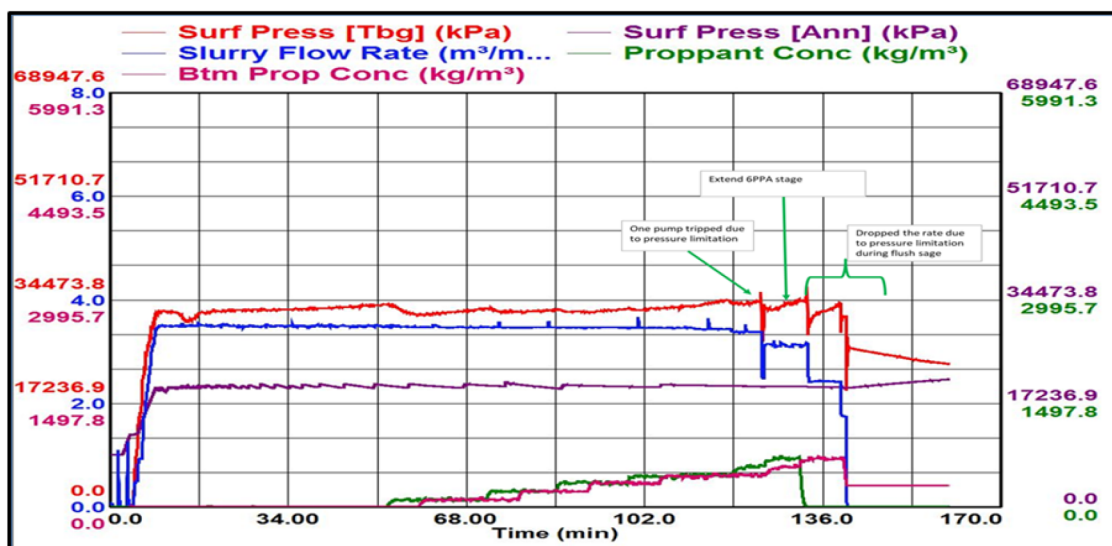


Figure 5—Main Frac treatment (A-6)

Post-fracture multi-rate testing yielded substantial improvement: the first test rate achieved 297 m³/d at a tubing head pressure (THP) of 3,200 kPa, while the second test rate after choking back the well was 158 m³/d at a THP of 5,500 kPa. The well is expected to deliver up to 340 m³/d at full opening (THP of 1,800 kPa), though production will be managed at 158 m³/d in line with reservoir management guidelines. Figure (6) shows well potential and testing post frac treatment.

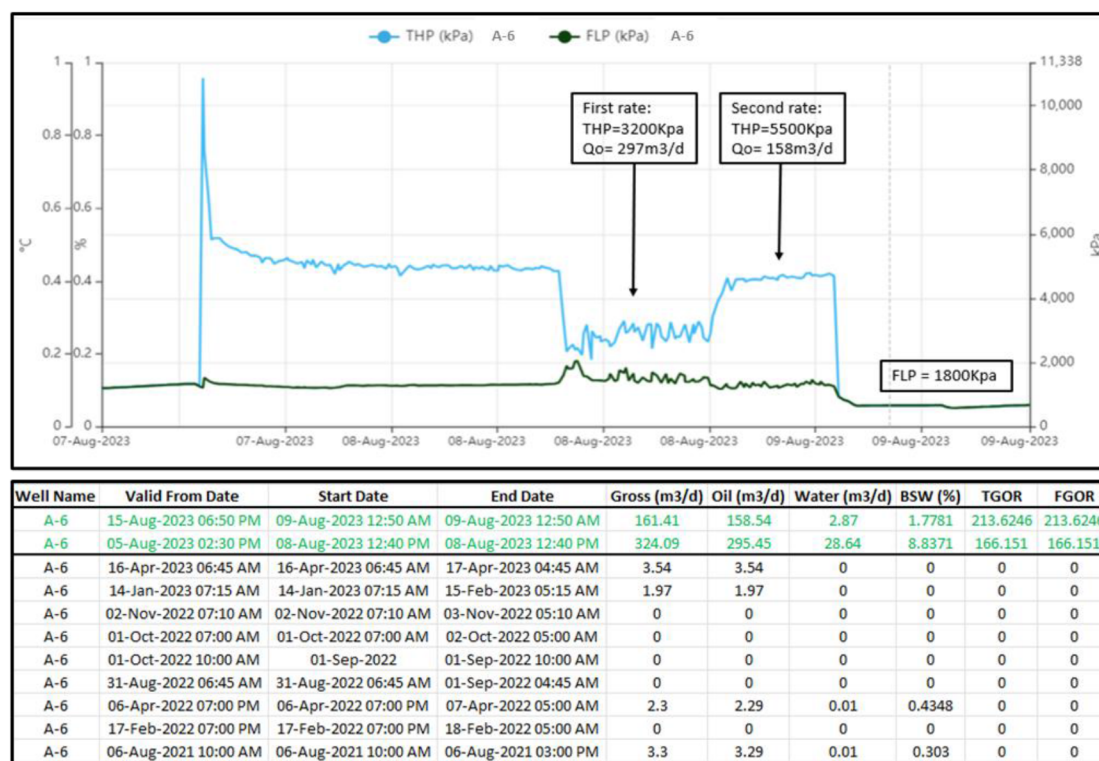


Figure 6—Post Frac potential and Testing Results (A-6)

Operationally, the fracturing program delivered strong economic benefits by avoiding two hoist workovers—one for pulling out existing completion and running the frac string, and another for converting the frac string to a production completion—resulting in cost savings of approximately USD 1.2 million. Additionally, the fracturing unit was efficiently utilized during downtime caused by issues at other wells, enabling the team to capitalize on available resources and avoid further deferment. Lessons learned from this operation highlighted the importance of enhanced coordination between engineering, operations, and subsurface teams to optimize post-frac suspension and re-hooking activities for future wells.

Case-2: Reviving a Long-Term Shut-In Producer

Well A-20 was originally drilled in November 2021 as a natural-flow oil producer targeting the UG-1 and UG-2 formations. Despite initial expectations, the well exhibited low productivity, likely due to skin effects from drilling and completion. An acid stimulation via coiled tubing (CTU) in October 2022 failed to enhance flow. Subsequent reperforation in May 2023 temporarily restored production to 21 m³/d (132 bbl/day), which quickly declined to zero. The well remained shut-in for over a year and was classified as a Closed-In Well (CIW).



Figure 7—Well A-20 performance prior Frac Treatment

Following the successful hydraulic fracturing of other wells in Field A, well A-20 was selected for stimulation. The initial attempt to frac the well failed due to reaching the maximum injection pressure. The team then thoroughly reviewed the pressure limitation to identify opportunities for safely pushing the pressure envelope. The TRSSSV pressure rating was the limiting factor which was reassessed through integrated technical analysis and risk assessment which helped to increase the execution envelope and was successfully implemented in September 2024. The details of this can be referred in sections below. The operation revived the well from a non-flowing status, with post-fracture test through a separator indicating production of 250 m³/d (1570 bbl/day) of net oil at 2,800 kPa THP. The well was reclassified as self-flowing and commercially viable, marking a significant turnaround from its prior CIW status.

Replace table below with manual table without Tib020 mentioned both prior and after frac test

Prior Frac Tests

Well Name	Valid From Date			Gross (m3/d)	Oil (m3/d)	Water (m3/d) ≡	BSW (%)	FGOR	Gas Out (m3/d)
A-20	21-Jun-2023 06:55 PM	2...	2...	2164	2162	0.02	0.0924	0	0
A-20	03-Nov-2022 10:30 AM	0...	0...	0	0	0	0	0	43.67
A-20	01-Mar-2022 10:00 AM	0...	0...	30	25	5	16.6667	40	1000

After Frac Test

Well: A-20

Formation:

Top interval :

Bottom interval :

Cumulative oil

Cumulative water

Cumulative gas

Sheet start date

Formation: 0.00

0.00 m3

0.0 m3

0 m3

26-Sep-24

		Rates (at Standard Conditions)							Fluids			
Date	Time hh:mm	BHP kPa	BHT deg C	Gross m3/d	Oil m3/d	Water m3/d	Gas m3/d	Oil SG	BSW %	CGR m3/mm m3	GOR m3/m3	WGR m3/mm m3
26-Sep	17:00			397.67	240.01	157.66	26.578	0.823	40	9030.5	111	5.932
26-Sep	17:30			395.84	239.78	156.06	27.442	0.823	39	8737.7	114	5.687
26-Sep	18:00			397.74	240.08	157.66	28.688	0.823	40	8368.8	119	5.496
26-Sep	18:30			394.47	237.49	156.98	29.298	0.823	40	8106.1	123	5.358
26-Sep	19:00			395.07	235.73	159.34	29.711	0.823	40	7934.2	126	5.363
26-Sep	19:30			404.61	245.27	159.34	29.651	0.823	39	8272.0	121	5.374
26-Sep	20:00			396.11	239.36	156.75	28.281	0.823	40	8463.7	118	5.543
26-Sep	20:30			399.64	242.07	157.57	28.318	0.823	39	8548.4	117	5.564
26-Sep	21:00			434.66	268.03	166.63	29.796	0.823	38	8995.4	111	5.592
26-Sep	21:30											

Figure 8—Well A-20 Testing results prior & post Frac treatment

Operational excellence was once again demonstrated by avoiding two hoist workovers achieving cost savings estimated at USD 1.2 million. The success of this operation validated the broader fracturing

campaign and demonstrated the effectiveness of applying lessons learned from earlier wells, such as Well A-6, to reduce design complexity and optimize resource deployment.

Case-3: New oil wells with zero rates to magnificent production

Wells A-29 and A-30 are newly commissioned oil wells in Field A, both wells were not able to produce anything -zero oil/water- although CTU N2 lifting were performed to support lifting the heavy drilling completion brine with no success. A comprehensive integrated multidisciplinary review was conducted, focusing on the hydraulic fracturing treatments applied to Gharif formation. Building on the success of hydraulic fracturing campaigns in Wells A-6 and A-12 in the previous year, fracture stimulation was planned for wells A-29 and A-30 to enhance production.

Due to a known shallow oil-water contact (OWC) in A-29, there was a significant risk that the fracturing treatment could extend into the water zone, potentially necessitating the installation of an electrical submersible pump (ESP) to sustain production. Historically, ESP performance in Field A wells has been suboptimal, largely due to inflow distribution and ESP gas-lock issues. Consequently, the fracturing design aimed to improve reservoir inflow characteristics to optimize ESP operational efficiency. Both these wells had similar limitations in the maximum surface injection pressure like Well A-20 due to TRSSSV and packer rating. Again, the integrated risk assessment and technical analysis was used to push the pressure envelope to safely execute hydraulic fracturing these wells.

The hydraulic fracturing operation for well A-30 was successfully executed, resulting in well flowback with an initial net oil production rate of 334 m³/day (2097 bbl/day). This positive outcome contrasts with well A-29, where production remains below expectations – although the inflow was enhanced and was confirmed, however with high BSW and not able to flow as self-flow-, highlighting the complexities associated with shallow OWC and the challenges in ESP deployment across overall field.

Cases Summary

The integrated hydraulic fracturing campaigns across the A- field have demonstrated the transformative potential of stimulation treatments in low-productivity, shut-in, and newly completed wells. The well A-6 fracture program elevated a marginal producer to a high-rate well while delivering substantial cost savings through operational efficiencies. The revival of A-20 from shut-in status further validates the effectiveness of fracture stimulation coupled with optimized operational execution.

The wells A-29 and A-30 cases emphasize the critical need for tailored fracture designs in reservoirs with shallow oil-water contacts and known artificial lift constraints. While A-30 responded positively with significant initial production, the challenges encountered in A-29 underscore the complexities of water management and ESP performance.

Collectively, these case studies highlight the value of rigorous fracture design, multidisciplinary collaboration, and adaptive operational strategies to unlock reserves and enhance field development economics. Future efforts will focus on refining stimulation approaches in complex reservoirs, optimizing artificial lift integration, and scaling these practices across the broader Field A to maximize hydrocarbon recovery.

Technical Challenges: Tubing-Retrieveable Safety Valve (TRSCSSSV) Pressure Constraints and Design Considerations

One of the critical technical challenges encountered during the hydraulic fracturing campaign in Field A was managing pressure limitations imposed by pre-installed tubing-retrieveable surface-controlled subsurface safety valves (TRSCSSSVs). These valves, originally designed for routine production scenarios, were not rated for the elevated pressures required during stimulation operations.

The original well completion strategy was not anticipating the requirement of hydraulic fracturing operations hence it was designed to a maximum surface pressure of 5,000 psi (34,470 kPa) to only serve wells integrity, aligning with the rated working pressure of the TRSCSSSVs installed in several wells. However, during project planning and treatment simulation, it became evident that effective fracture propagation—particularly in deeper, more compacted intervals—required higher injection pressures. Consequently, fracturing designs in select wells were revised to accommodate surface pressures up to 5,800 psi (40,000 kPa).

This decision introduced a risk of exceeding the TRSCSSSV pressure envelope, which could compromise valve integrity or fail to maintain open functionality under elevated differential pressures. To address this, detailed pressure analysis and valve behavior modeling were conducted using standard valve control logic and safety design equations.

TRSCSSSV Hold-Open Pressure Calculation

The hold-open pressure (P_{so})—defined as the pressure required in the control line to keep the safety valve open during high-pressure operations—is governed by the following relationship:

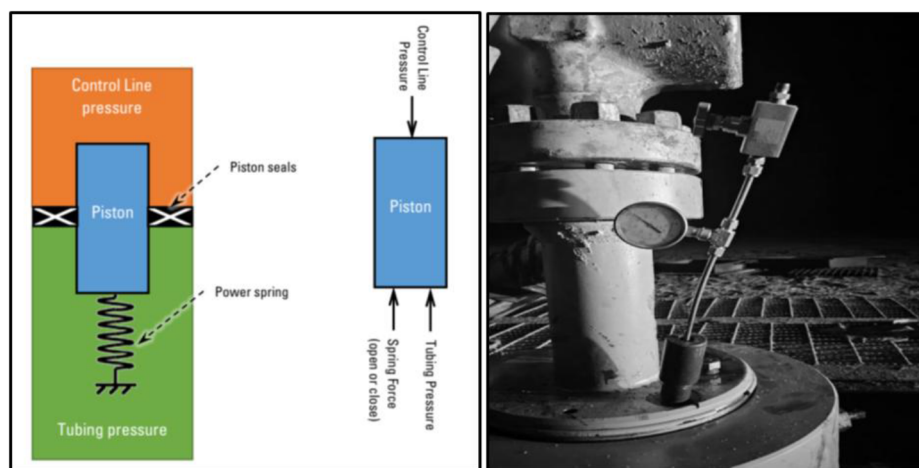


Figure 9—TRSCSSSV Hold Open Pressure, functionality and on Job Monitoring

$$P_{so} = P_t + P_o + P_s - P_h$$

Where:

- P_{so} = Hold-Open Surface Operating Pressure (psi)
- P_t = Tubing pressure at safety valve setting depth (psi)
- P_o = Opening pressure (psi) = $1.2 \times P_o$ engineering based on OneCAT or vendor specifications
- P_s = Safety factor, typically not less than 400 psi
- P_h = Hydrostatic pressure of the hydraulic control fluid in the control line (psi)

This calculation helps define the pressure window within which the safety valve will remain reliably open, preventing premature closure due to pressure imbalance or insufficient hydraulic control force.

Operational Risk and Mitigation

Given that the TRSCSSSVs were only rated for 5,000 psi, fracturing operations requiring pressures up to 5,800 psi posed a risk of:

- Valve seat leakage or mechanical failure due to overpressure,
- Uncommanded valve closure, interrupting treatment,
- Loss of tubing integrity at the valve location.

To mitigate these risks, the following strategies were employed:

- Deep assessment with the OEM and service provider of the TRSSSV engineering design showed that it was designed for much higher pressure than the nameplate rating (not tested).
- The expected pressure during the frac treatment was much below the design limits.
- Assessing the TRSSSV behavior during the Mini Frac prior to executing the main treatment planned to increase confidence in safe execution
- Real-time surface pressure monitoring with automated shutdown protocols.
- Contingency planning, including a conservative fallback option to cap treatments at 5,000 psi if early signs of valve instability were observed.

Despite operating slightly above rated pressure in selected wells, no valve failures were observed, validating the robustness of the control systems and confirming the reliability of the calculated hold-open pressure envelope.

The experience also underlined the importance of aligning safety valve specifications with future well stimulation needs during initial completion planning. In similar fields, proactive design adjustments—such as high-pressure TRSCSSSVs and packers—may be considered to expand operational flexibility without compromising safety.

Conclusions and Recommendations

The hydraulic fracturing campaign executed in Field A represents a breakthrough in the development of deep, low-permeability clastic reservoirs under challenging economic and operational conditions. The successful stimulation of five wells led to a dramatic uplift in production, with total output increasing from 44 bbl/day to over 4,088 bbl/day of net oil. The project delivered an estimated NPV of \$158 million, while achieving a Unit Technical Cost of just \$1.04/bbl — setting a new benchmark for cost-effective reservoir stimulation in the region. [Table \(1\)](#) indicates production results pre/post Fracturing treatment.

This success was made possible through a multidisciplinary and integrated approach that emphasized:

- Cost optimization, by leveraging existing completions and avoiding hoist-based workovers
- Safe extension of operating envelopes, through rigorous modeling, risk assessments, and real-time monitoring
- Operational efficiency, achieved by continuous learning and iterative design improvements
- Environmental responsibility, through reduced GHG emissions and elimination of diesel-powered artificial lift

These results validate the feasibility and scalability of applying hydraulic fracturing in mature or underperforming reservoirs that traditionally faced economic and technical barriers to stimulation. Furthermore, the campaign demonstrates that pushing beyond conventional operating constraints, when done safely and systematically, can unlock significant value without compromising well integrity or safety.

Table 1—Production Results on the Five Fracced Wells Pre/Post treatments

Well	Pre-Frac Oil (m3/d)	Post Frac Oil (m3/d)
A-6	2	+275
A-12	2	+30 (Required an ESP)
A-30	0 (New oil)	334
A-24	3	+80
A-20	0 (CIW)	+250
Total	7 m3/d (~7 bbl/day)	~690 m3/d (~4300 bbl/day)

Recommendations

Based on the findings of this study, the following recommendations are proposed for operators considering similar hydraulic fracturing applications:

1. Prioritize candidate selection using robust diagnostic workflows, including reservoir modeling, production analysis, and well integrity assessment, to ensure high-impact stimulation opportunities.
2. Leverage existing well infrastructure wherever feasible but ensure detailed engineering evaluations are conducted to determine safe operating margins.
3. Implement integrated risk assessments early in the planning phase to identify potential equipment limitations and operational hazards.
4. Adopt a phased implementation strategy, beginning with pilot wells to establish performance baselines and inform field-wide scaling.
5. Invest in real-time data acquisition and monitoring systems, which are essential for validating fracture execution, ensuring safety, and enabling rapid operational adjustments.
6. Consider environmental and sustainability metrics as key performance indicators in hydraulic fracturing projects, especially in regions with strict ESG mandates.

This case study from Field A serves as a replicable model for sustainable, innovative, and economically viable hydraulic fracturing in challenging environments. The methodology, results, and lessons learned from this initiative can be applied to similar reservoirs across the region and globally, supporting long-term energy resilience and responsible hydrocarbon development.

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Nomenclature

bbbl Barrel (of crude oil)
ESP Electric Submersible Pump

GHG	Greenhouse Gas
HSE	Health, Safety, and Environment
IRA	Integrated Risk Assessment
NPV	Net Present Value
SPE	Society of Petroleum Engineers
UTC	Unit Technical Cost
psi	Pounds per square inch
frac	Fracture / Fracturing
MAASP	Maximum Allowable Annulus Surface Pressure
SIT	Sub-Surface Integrity Test
WIT	Well Integrity Test
GOFFER	powerful hydraulic fracturing simulation software
TRSCSSSV	Tubing-Retrievable Surface Control Sub-Surface Safety Valve
RIH	Run in Hole
CiW	Closed-In Well
FDP	Field Development Plan
CTU	Coiled Tubing Unit
RDL	Reservoir Data Log
STOIIP	Stock Tank Oil Initially In Place
OEM	Original Equipment Manufacturer